# CLOUD COMPUTING AND BIG DATA ANALYTICS

## CSA15 – Assignment Answer

### Design and Deployment of a Cloud-Based Virtualized Collaboration Platform for an Engineering College

| Student Details | Value |
|---|---|
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| COURSE CODE | CSA1507 |
| COURSE NAME | Cloud Computing and Big Data Analytics |
| CO | CO1, CO2, CO3 |
| SDG Mapping | SDG 4 – Quality Education; SDG 8 – Decent Work and Economic Growth; SDG 9 – Industry, Innovation and Infrastructure |

## 1. Problem Analysis

A small engineering college currently operates separate physical systems for application hosting, file storage, and user access. This leads to high infrastructure cost, inefficient resource utilization, difficult capacity management during peak demand, and additional administrative overhead. The proposed solution is a cloud-based virtualized environment that combines infrastructure virtualization, cloud service models, secure remote access, and collaboration services.

### 1.1 Institutional Requirements

| Requirement | Proposed Cloud/Virtualized Solution | Reason |
|---|---|---|
| Online learning | LMS/application VM with web access | Supports faculty and students through browsers and remote access. |
| File sharing | Dedicated file-storage VM / cloud storage | Centralized storage with controlled access. |
| Application hosting | Application server VM / PaaS | Separates application deployment from physical hardware. |
| User authentication | Identity/access service | Provides controlled access based on user roles. |
| Collaborative editing | SaaS collaboration suite | Enables simultaneous document editing and sharing. |
| Communication | Cloud email/chat/video-conferencing service | Supports academic and administrative communication. |
| Scalability | Virtual CPU/RAM/storage allocation and cloud scaling | Resources can be increased when demand rises. |
| Security | Firewall, IAM, least privilege, encryption, backups | Protects institutional data and services. |

## 2. Cloud Service Model Identification and Requirement Analysis

The proposed platform uses IaaS, PaaS, and SaaS together. Each model is assigned to the type of responsibility that it handles most effectively.

| Service Model | Example in Proposed Platform | Institution Responsibility | Main Use |
|---|---|---|---|
| IaaS | Virtual machines, virtual network, storage, firewall | Configure OS, VM resources, applications and access policies | Infrastructure services and VM-based data center |
| PaaS | Managed application/database/runtime platform | Develop, configure and maintain application code | Application deployment without managing underlying servers |
| SaaS | Cloud office suite, online storage, communication and collaboration tools | Manage users, permissions and content | Document sharing, editing, communication and productivity |

**2.1 Analysis of IaaS**

- IaaS provides virtualized compute, storage and networking resources while allowing the institution to control the operating systems and deployed services.
- It is suitable for the virtualized data-center demonstration because separate VMs can represent the LMS, file server, application server and monitoring/administration services.
- Scalability is achieved by changing VM CPU, memory and storage allocations or creating additional instances.
- The institution pays for allocated/consumed infrastructure rather than purchasing a separate physical server for every service.
- Security responsibilities remain significant because the institution must configure operating-system security, user permissions, firewall rules, patching and application security.

**2.2 Analysis of PaaS**

- PaaS provides a managed application runtime, reducing the need to administer the underlying operating system and server infrastructure.
- It is appropriate for college-developed portals, attendance applications, departmental applications and APIs.
- PaaS improves developer productivity because deployment, runtime management and scaling are handled by the platform.
- The major limitation is reduced infrastructure-level control and possible vendor/platform dependency.

- Security still requires secure application code, identity management, secrets management and correct configuration.

### 2.3 Analysis of SaaS

- SaaS delivers complete applications through the network and is the most appropriate model for collaboration activities.
- Examples include browser-based document editing, file sharing, email, chat, calendars and online meetings.
- SaaS minimizes installation and maintenance work for the college IT team.
- Accessibility is high because users can work from campus or remotely using supported devices.
- The main limitations are dependency on the service provider, network availability, subscription cost and reduced control over the underlying application platform.

### 2.4 Overall Service Model Comparison

| Aspect | IaaS | PaaS | SaaS |
|---|---|---|---|
| Control | High | Medium | Low |
| Administration effort | High | Medium | Low |
| Scalability | High | High | Provider-managed |
| Typical college use | VMs, storage, networks | Application development | Collaboration/productivity |
| Cost model | Resource-based | Platform/usage-based | Subscription/usage-based |
| Security responsibility | Shared; institution manages more layers | Shared application/platform responsibility | Mostly provider-managed infrastructure; institution manages users/data |

## 3. Virtualized Data Center Architecture

A small virtualized data center is proposed using a Type-2 virtualization environment such as VirtualBox for laboratory demonstration. The same logical design can be mapped to an IaaS provider for production use.

### 3.1 Logical Architecture

| Layer / Component | Connectivity | Layer / Component |
|---|---|---|
| Students / Faculty / Admin | Internet / Campus Network | Secure Gateway / Firewall |
| | ↓ | |
| | Virtualization Host | |

| | | |
|---|---|---|
| VM1 – LMS/Web Server | VM2 – File Server | VM3 – Application Server |
| VM4 – Database Server | VM5 – Monitoring/Admin | Virtual Network / Internal LAN |
| | ↓ | |
| Cloud SaaS Collaboration: Documents + File Sharing + Communication | | |

### 3.2 VM Design

| VM | Service | vCPU | RAM | Storage | Network |
|---|---|---|---|---|---|
| VM1 | LMS / Web Server | 2 | 4 GB | 40 GB | Bridged + Internal |
| VM2 | File Server | 2 | 4 GB | 100 GB | Bridged + Internal |
| VM3 | Application Server | 2 | 4 GB | 50 GB | Internal |
| VM4 | Database Server | 2 | 4 GB | 60 GB | Internal |
| VM5 | Monitoring / Administration | 1 | 2 GB | 30 GB | Bridged + Internal |

The exact allocation can be adjusted according to the host computer's available resources. For a classroom demonstration, the VMs can be started selectively rather than running every VM simultaneously.

## 4. Virtual Machine Deployment and Resource Configuration

The implementation can be performed in a controlled laboratory environment. The following procedure provides reproducible deployment steps and the evidence that should be captured in screenshots.

1. Install a virtualization tool such as VirtualBox on the host computer.
2. Create an internal virtual network for service-to-service communication.
3. Create VM1 for the LMS/Web service and install a Linux server operating system.
4. Create VM2 for file storage and configure a shared directory with user permissions.
5. Create VM3 for the college application service.

6. Create VM4 for the database service and restrict database access to the application network.
7. Create VM5 for administration/monitoring and secure remote management.
8. Assign CPU, memory, disk and network resources according to the VM design table.
9. Configure static or reserved IP addresses for infrastructure services.
10. Configure firewall rules so only required ports are exposed.
11. Test service-to-service communication using ping, SSH and application/database connection tests.
12. Capture screenshots of VM creation, CPU/RAM settings, storage settings, network settings and successful service access.

### 4.1 Example Resource Configuration

Host Network: Campus/Internet connection
Internal Network: CloudLab-Internal
VM1 IP: 192.168.50.11 -> Web/LMS
VM2 IP: 192.168.50.12 -> File Server
VM3 IP: 192.168.50.13 -> Application
VM4 IP: 192.168.50.14 -> Database
VM5 IP: 192.168.50.15 -> Administration

## 5. Network Configuration and Service Communication

The network is divided logically so that public-facing services and internal services do not have identical exposure. The web/LMS endpoint can be reachable from users, while the database and internal file services should remain restricted.

| Source | Destination | Purpose | Security Rule |
|---|---|---|---|
| Student/Faculty | VM1 | Access LMS/Web | Allow HTTPS; redirect HTTP to HTTPS where appropriate |
| Student/Faculty | VM2 | Access permitted files | Allow only authenticated file access |
| VM3 | VM4 | Application database access | Allow database port only from VM3 |
| Admin | VM1–VM5 | Remote administration | Allow SSH/RDP only to authorized administrators |
| All VMs | Monitoring VM | Logs/monitoring | Allow required monitoring traffic |

### 5.1 Connectivity Test

1. From VM3, ping VM4.
2. From VM3, test the database service on VM4.
3. From a client system, open the LMS URL hosted by VM1.
4. From an authorized account, access the shared files on VM2.
5. Attempt an unauthorized database connection from a student/client machine.
6. Confirm that the firewall blocks the unauthorized request.

## 6. Cloud Access and Remote Resource Management

Secure remote access is implemented using authenticated administrative access. SSH can be used for Linux VM administration, while browser-based HTTPS is used for user-facing applications. Administrative access should be restricted by firewall rules and strong authentication.

- Use HTTPS for browser-based services to protect credentials and data in transit.
- Use SSH keys rather than password-only administration where supported.
- Restrict management ports to trusted administrator networks or a VPN.
- Use multi-factor authentication for cloud/SaaS accounts whenever available.
- Maintain audit logs for authentication, file access and administrative actions.
- Back up critical configuration and academic data regularly.

## 7. User Roles, Permissions and Security

| Role | Permissions | Restrictions |
| --- | --- | --- |
| Administrator | Manage VMs, network, users, backups and security policies | Administrative access only for authorized IT staff |
| Faculty | Create/edit course materials, upload files, collaborate | Cannot change infrastructure or other users' permissions |
| Student | View/download permitted material, submit assignments, collaborate | Read/submit access; no administrative privileges |
| Guest | Limited public information | No access to private academic files or internal services |

### 7.1 Security Controls

- Least-privilege access: users receive only the permissions required for their academic role.

- Network segmentation: database and management services remain on the internal network.
- Encryption: HTTPS/TLS protects web traffic; encrypted storage/backups should be used for sensitive data.
- Authentication: unique accounts, strong passwords and MFA for privileged/cloud accounts.
- Backup and recovery: scheduled backups protect against accidental deletion and service failure.
- Patching: operating systems and applications must be updated regularly.
- Monitoring: logs should be reviewed for failed logins, unusual access and service failures.

## 8. Cloud-Based Collaboration Services

| Activity | Cloud Collaboration Function | Expected Benefit |
|---|---|---|
| Document sharing | Central cloud folder with role-based access | Single source of truth for academic documents |
| Collaborative editing | Multiple users edit shared documents | Reduces duplicate document versions |
| File access | Browser/mobile/desktop access to authorized files | Improves accessibility |
| Communication | Email, chat and online meetings | Supports remote academic coordination |
| Assignment submission | Controlled student submission folders | Simplifies collection and review |
| Department collaboration | Shared calendars and team workspaces | Improves coordination and scheduling |

## 9. Comparison with Traditional Infrastructure

| Aspect | Traditional Physical Infrastructure | Proposed Cloud/Virtualized Infrastructure |
|---|---|---|
| Resource utilization | Often low because servers are dedicated to individual services | Higher through VM consolidation and shared resources |
| Scalability | Requires purchasing/installing additional hardware | Resources can be resized or additional instances created |
| Accessibility | Often dependent on campus network/VPN setup | Services can be securely accessed remotely |
| Cost | High upfront hardware and | Lower upfront cost; recurring |

| | | |
|---|---|---|
| | maintenance cost | resource/service cost |
| Management | Physical hardware and OS management | Centralized virtualization and cloud management |
| Security | Institution controls physical environment but must manage all security layers | Shared-responsibility model; requires correct IAM and configuration |
| Collaboration | File copies and local systems can create version conflicts | Centralized SaaS collaboration supports simultaneous work |
| Availability | Hardware failure may interrupt a service | Virtualization, backups and provider capabilities can improve resilience |

### 9.1 Advantages

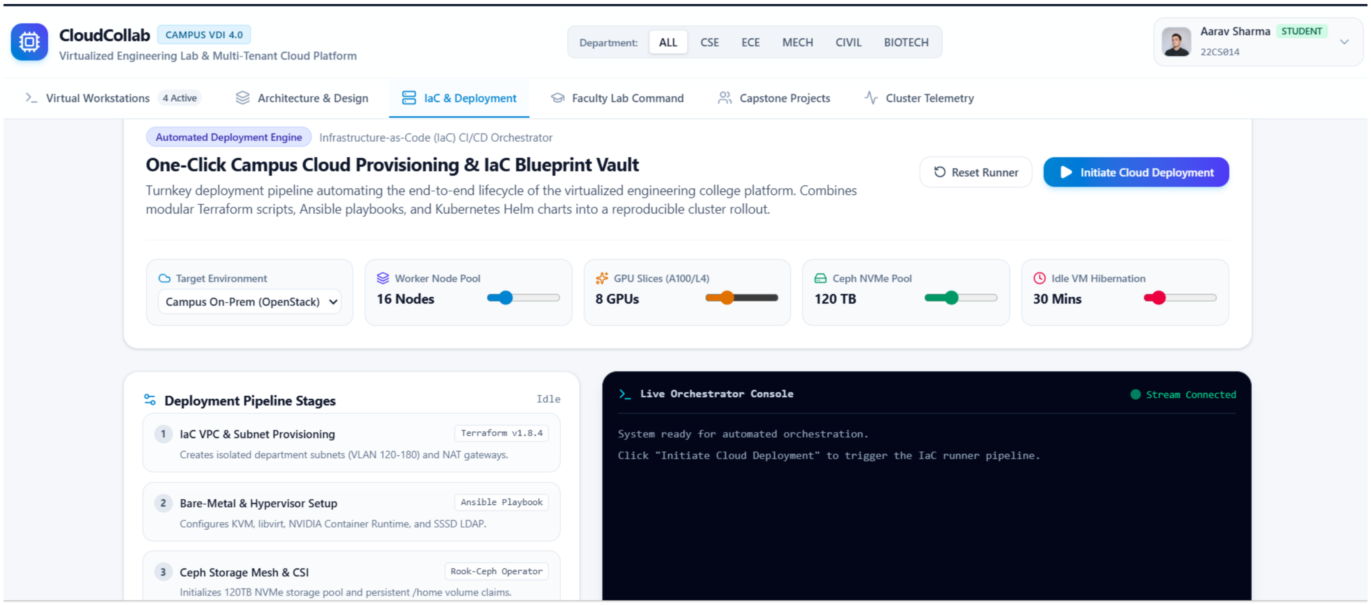
- Better resource utilization through virtualization.
- Rapid provisioning of new services.
- Improved scalability during examination or admission peaks.
- Remote access supports online learning and hybrid academic work.
- Centralized collaboration reduces duplicate files and version conflicts.
- Infrastructure can be managed from a smaller number of administrative interfaces.

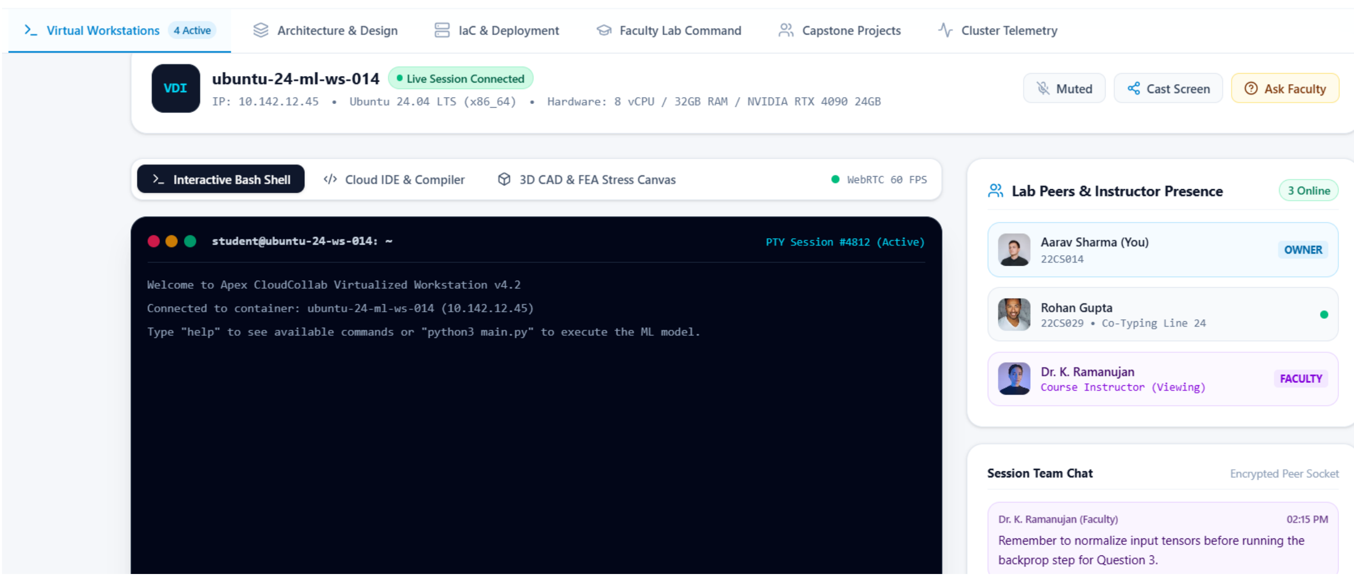
### 9.2 Limitations and Challenges

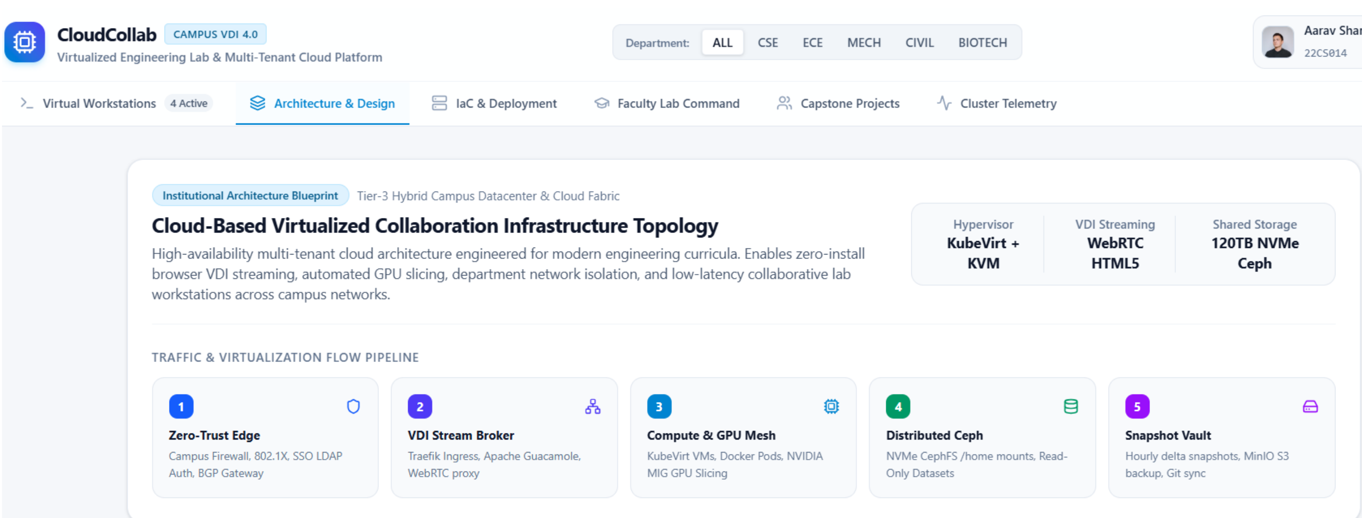
- Dependence on reliable network connectivity.
- Cloud subscription and usage costs must be monitored.
- Misconfigured identity, storage or firewall rules can create security exposure.
- Vendor lock-in may make migration difficult.
- Virtualized environments require careful CPU, memory and storage planning.
- Large-scale cloud deployments require skilled administration and monitoring.

## 10. Implementation Results and Validation Plan

The following table defines the expected evidence for the completed laboratory implementation. Actual screenshots should be inserted after deployment; this report does not fabricate execution evidence.


CloudCollab
CAMPUS VDI 4.0
Virtualized Engineering Lab & Multi-Tenant Cloud Platform
Department: ALL CSE ECE MECH CIVIL BIOTECH
Aarav Sharma STUDENT
22CS014
Virtual Workstations 4 Active
Architecture & Design
IaC & Deployment
Faculty Lab Command
Capstone Projects
Cluster Telemetry
Automated Deployment Engine
Infrastructure-as-Code (IaC) CI/CD Orchestrator
One-Click Campus Cloud Provisioning & IaC Blueprint Vault
Turnkey deployment pipeline automating the end-to-end lifecycle of the virtualized engineering college platform. Combines modular Terraform scripts, Ansible playbooks, and Kubernetes Helm charts into a reproducible cluster rollout.
Reset Runner
Initiate Cloud Deployment
Target Environment
Campus On-Prem (OpenStack)
Worker Node Pool
16 Nodes
GPU Slices (A100/L4)
8 GPUs
Ceph NVMe Pool
120 TB
Idle VM Hibernation
30 Mins
Deployment Pipeline Stages
Idle
1 IaC VPC & Subnet Provisioning
Terraform v1.8.4
Creates isolated department subnets (VLAN 120-180) and NAT gateways.
2 Bare-Metal & Hypervisor Setup
Ansible Playbook
Configures KVM, libvirt, NVIDIA Container Runtime, and SSSD LDAP.
3 Ceph Storage Mesh & CSI
Rook-Ceph Operator
Initializes 120TB NVMe storage pool and persistent /home volume claims.
Live Orchestrator Console
Stream Connected
System ready for automated orchestration.
Click "Initiate Cloud Deployment" to trigger the IaC runner pipeline.


Virtual Workstations 4 Active
Architecture & Design
IaC & Deployment
Faculty Lab Command
Capstone Projects
Cluster Telemetry
VDI
ubuntu-24-ml-ws-014
Live Session Connected
IP: 10.142.12.45 • Ubuntu 24.04 LTS (x86_64) • Hardware: 8 vCPU / 32GB RAM / NVIDIA RTX 4090 24GB
Muted
Cast Screen
Ask Faculty
Interactive Bash Shell
Cloud IDE & Compiler
3D CAD & FEA Stress Canvas
WebRTC 60 FPS
student@ubuntu-24-ws-014: ~
PTY Session #4812 (Active)
Welcome to Apex CloudCollab Virtualized Workstation v4.2
Connected to container: ubuntu-24-ml-ws-014 (10.142.12.45)
Type "help" to see available commands or "python3 main.py" to execute the ML model.
Lab Peers & Instructor Presence
3 Online
Aarav Sharma (You)
22CS014
OWNER
Rohan Gupta
22CS029 • Co-Typing Line 24
Dr. K. Ramanujan
Course Instructor (Viewing)
FACULTY
Session Team Chat
Encrypted Peer Socket
Dr. K. Ramanujan (Faculty)
02:15 PM
Remember to normalize input tensors before running the backprop step for Question 3.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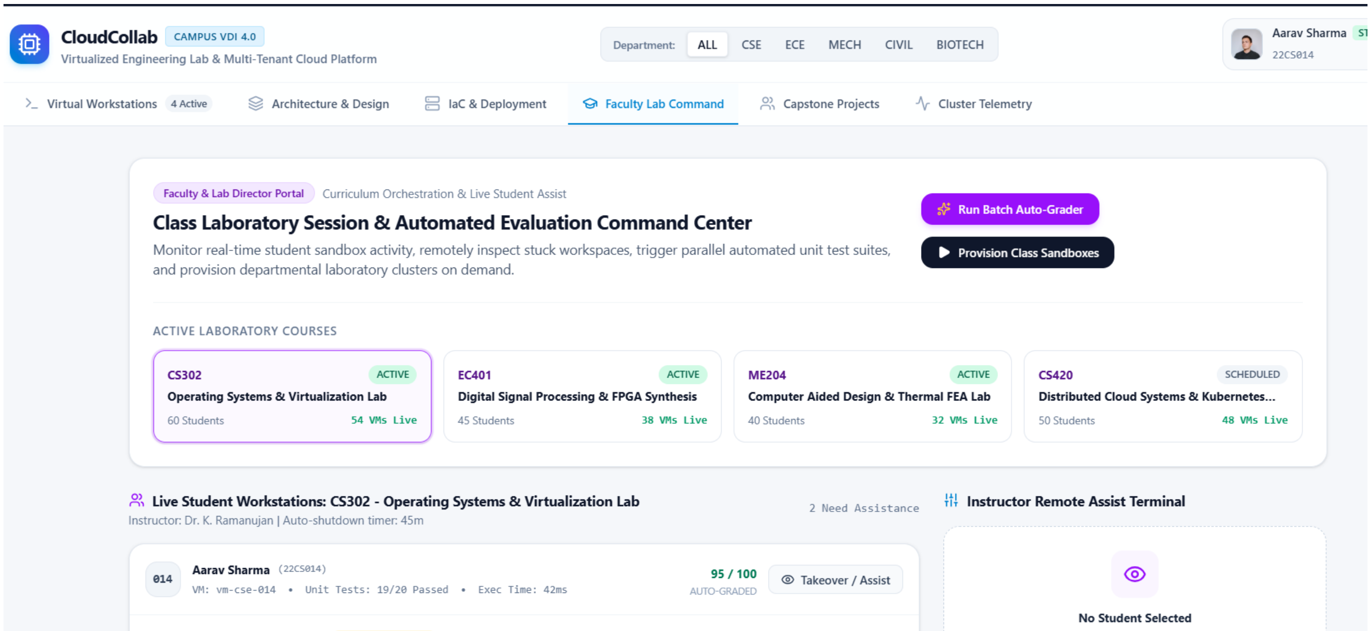
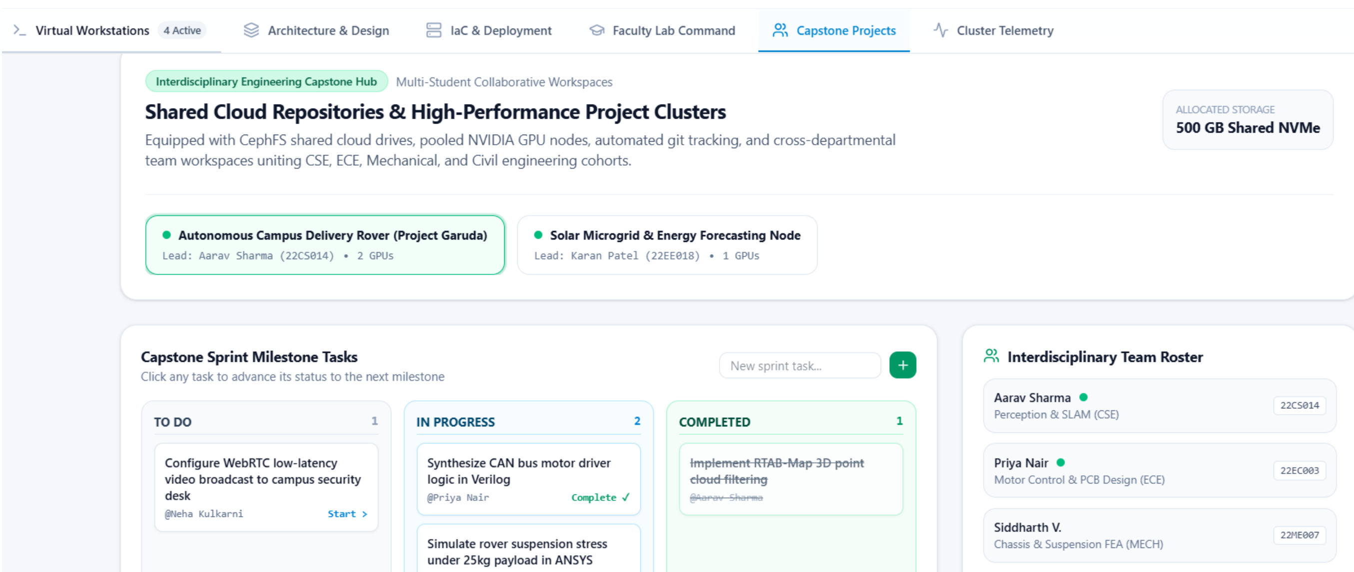
Capstone Projects
Cluster Telemetry
Institutional Architecture Blueprint
Tier-3 Hybrid Campus Datacenter & Cloud Fabric
Cloud-Based Virtualized Collaboration Infrastructure Topology
High-availability multi-tenant cloud architecture engineered for modern engineering curricula. Enables zero-install browser VDI streaming, automated GPU slicing, department network isolation, and low-latency collaborative lab workstations across campus networks.
Hypervisor
KubeVirt + KVM
VDI Streaming
WebRTC HTML5
Shared Storage
120TB NVMe Ceph
TRAFFIC & VIRTUALIZATION FLOW PIPELINE
1 Zero-Trust Edge
Campus Firewall, 802.1X, SSO LDAP Auth, BGP Gateway
2 VDI Stream Broker
Traefik Ingress, Apache Guacamole, WebRTC proxy
3 Compute & GPU Mesh
KubeVirt VMs, Docker Pods, NVIDIA MIG GPU Slicing
4 Distributed Ceph
NVMe CephFS /home mounts, Read-Only Datasets
5 Snapshot Vault
Hourly delta snapshots, MinIO S3 backup, Git sync




| Test | Expected Result | Evidence to Capture |
|---|---|---|
| VM creation | Five VMs visible in the virtualization manager | VM manager screenshot |
| Resource configuration | CPU/RAM/storage match the design | Settings screenshots |
| Network connectivity | Required VMs communicate successfully | Ping/network test screenshot |
| LMS/Web access | Authorized client opens the service | Browser screenshot |
| File access | Authorized users can access permitted files | File-service screenshot |
| Database security | Only application VM can access | Connection/firewall test |

| | | |
|---|---|---|
| | database | |
| Remote administration | Authorized admin can securely manage VMs | SSH/VPN/admin screenshot |
| Role-based access | Student/faculty/admin permissions differ | Permission test screenshots |
| Collaboration | Users can share/edit documents according to permissions | Collaboration-service screenshot |

## 11. Analysis of Scalability, Cost, Accessibility and Security

| Factor | Analysis |
|---|---|
| Resource utilization | Virtual machines allow multiple services to share one physical host. CPU and memory can be allocated according to workload. |
| Scalability | Cloud resources can be resized or replicated as demand increases. This is useful during admissions, examinations and assignment deadlines. |
| Accessibility | Browser-based services and secure remote access allow students and faculty to use services beyond the physical campus. |
| Cost | Virtualization reduces the number of dedicated physical servers. Cloud services change capital expenditure into operating/resource-based expenditure, but ongoing usage must be controlled. |
| Security | Identity, least privilege, segmentation, encryption, patching, backups and monitoring are required. Cloud adoption does not remove security responsibility. |
| Collaboration | Centralized SaaS tools improve simultaneous editing, sharing and communication compared with manually exchanging file copies. |

## 12. Deployment Algorithm / Pseudocode

```
CLOUD_VIRTUALIZED_COLLEGE_PLATFORM()
    analyze institutional requirements
    classify requirements as IaaS, PaaS or SaaS

    install virtualization platform
    create internal virtual network
```

```
create VM1 for LMS/Web
create VM2 for File Storage
create VM3 for Application Service
create VM4 for Database Service
create VM5 for Administration/Monitoring

assign CPU, RAM, storage and IP configuration
install required operating systems and services
configure firewall and network segmentation
configure users and role-based permissions
enable secure remote administration
configure collaboration services

test VM-to-VM communication
test application, file and database access
test unauthorized access restrictions
configure backup and monitoring

record screenshots and implementation results
evaluate scalability, cost, accessibility, security and collaboration
return final cloud platform design and results
```

**13. GitHub Upload Structure**

A clean repository can be organized as follows:

```
cloud-virtualized-collaboration-platform/
├── README.md
├── architecture/
│   └── architecture-diagram.png
├── virtualization/
│   ├── vm-configuration.md
│   └── network-configuration.md
├── deployment/
│   └── deployment-steps.md
├── security/
│   └── roles-and-permissions.md
├── screenshots/
│   ├── vm-manager.png
```

```
│   ├── network-test.png
│   └── service-access.png
└── report/
    └── cloud-assignment.docx
```

The README should contain the problem statement, architecture, software requirements, deployment steps, test results, security considerations and limitations. Screenshots should be original evidence from the actual implementation.

## 14. Final Conclusion

The proposed cloud-based virtualized collaboration platform addresses the institution's requirements for online learning, file sharing, application hosting, secure access and collaborative academic activities. IaaS is appropriate for the virtualized infrastructure and infrastructure services, PaaS is suitable for managed application development and deployment, and SaaS is appropriate for collaboration and productivity.

The virtualized data-center design uses multiple VMs with dedicated responsibilities while sharing the underlying physical resources. Network segmentation and role-based access control reduce unnecessary exposure, while HTTPS, secure remote administration, backups, patching and monitoring provide essential security controls.

Compared with traditional physical infrastructure, the proposed design improves resource utilization, scalability, accessibility and collaboration, while introducing new considerations such as network dependency, recurring cloud cost, vendor dependency and shared-responsibility security. The design therefore provides a practical foundation for an engineering college to migrate services incrementally to a cloud environment.

## 15. Expected Learning Outcomes

- CO1: Identify and analyze IaaS, PaaS and SaaS according to institutional requirements.
- CO2: Design and configure a virtualized data-center environment with multiple service VMs and network communication.
- CO3: Demonstrate secure cloud access, role-based permissions and cloud collaboration mechanisms.
- Understand the trade-offs among resource utilization, scalability, cost, accessibility, security and collaboration.

## 16. Reference Basis

This answer follows the organization and expected deliverables of the provided DAA assignment reference: problem analysis, algorithms/procedures, step-by-step implementation, results/evidence, complexity or analytical discussion, comparison, limitations, and final conclusion. The Cloud Computing assignment's stated deliverables are architecture design, virtualized data-center implementation/results, cloud service model analysis, VM evidence, comparison/analysis, and GitHub upload.